

Conjugate Priors

Math 742

2026-01-26

Intro

We learned last time that **Bayes** $\rightarrow$ updates prior beliefs with data.

Now we look at **conjugate priors**. Conjugate priors basically make that update look like adding information.

Example 1: Suppose we'd like to estimate the probability (p) of getting a head when flipping a coin. We're not sure if the coin is "fair" or "weighted".

Suppose in the past the coin was flipped twice. The result was 1 head and 1 tail. Therefore, I decide to model our prior belief with a Beta(2,2).

$$\pi(p) \propto p(1-p)$$

Not convinced that this is correct, we take some more data. This time we flip the coin 10 times and the result is 7 heads and 3 tails.

Combining the prior with the data we just collected, our new posterior probability density function is:

$$p \mid x \sim \text{Beta}(2 + 7, 2 + 3) = \beta(9, 5) = 6435x^8(1-x)^4$$

The posterior mean is:

$$\frac{\alpha}{\alpha + \beta} = \frac{9}{9 + 5} = 0.6429$$

The posterior mean is a compromise between the sample mean (7/10) and the prior mean (0.5).

Example 2:

Suppose I have a normal distribution, I think it's mean is 0, but I'm pretty uncertain about this.

$$X \mid \mu \sim N(\mu, \sigma^2), \text{ where } \sigma^2 \text{ is known}$$

and the prior for μ is

$$\mu \sim N(0, 4)$$

To be sure, I collect some data ($n = 20$) and it turns out the sample mean is 3 and variance 1.

For my posterior mean, I want to weight the data and the prior information. I want to do this in such a way that it accounts for the precision which is $1/\text{Variance}$.

The prior mean precision is $1/4 = 0.25$. The precision of the data is $20/1 = 20$

This shows that the posterior is very close to the sample mean because the data contains much more information than the prior.

So my weight should be,

$$w = \frac{\text{data precision}}{\text{data precision} + \text{prior precision}} = \frac{20}{20 + 1/4} = 0.9877$$

The posterior mean should thus be a weighted average of the prior mean and the data.

$$\begin{aligned}\mu_{new} &= 0.9877 \times \bar{x} + (1 - 0.9877) \times \mu_0 \\ &= 0.9877(3) + 0.0123 \times 0 \approx 2.96\end{aligned}$$

In both examples, the posterior is formed by adding prior information and data information.

Why Conjugacy??

Think of the *prior* as a container of a certain shape. The *likelihood* pours information into the container. A *conjugate prior* is chosen so that pouring in the data the container keeps the same shape just with new markings.

→ A conjugate prior is a prior whose shape is preserved after updating — the data just relabel the parameters.

Without conjugacy:

- normalization integrals hard
- simulation needed

With conjugacy:

- posterior known in closed form
- easy interpretation

Conjugacy works because prior and likelihood share the same functional form.

Definition 1: A prior is conjugate if the posterior belongs to the same parametric family as the prior.

Beta-Binomial Model

$$X|p \sim \text{Binomial}(n, p)$$

$$p \sim \text{Beta}(\alpha, \beta)$$

Likelihood:

$$p(x|p) \propto p^x (1-p)^{n-x}$$

Prior:

$$\pi(p) \propto p^{\alpha-1} (1-p)^{\beta-1}$$

Posterior:

$$f(p|x) \propto p^{\alpha+x-1} (1-p)^{\beta+n-x-1}$$

Interpretation

- $\alpha - 1$: prior successes
- $\beta - 1$: prior failures
- Posterior = prior + data

Posterior mean:

$$\mathbb{E}[p|x] = \frac{\alpha + x}{\alpha + \beta + n}$$

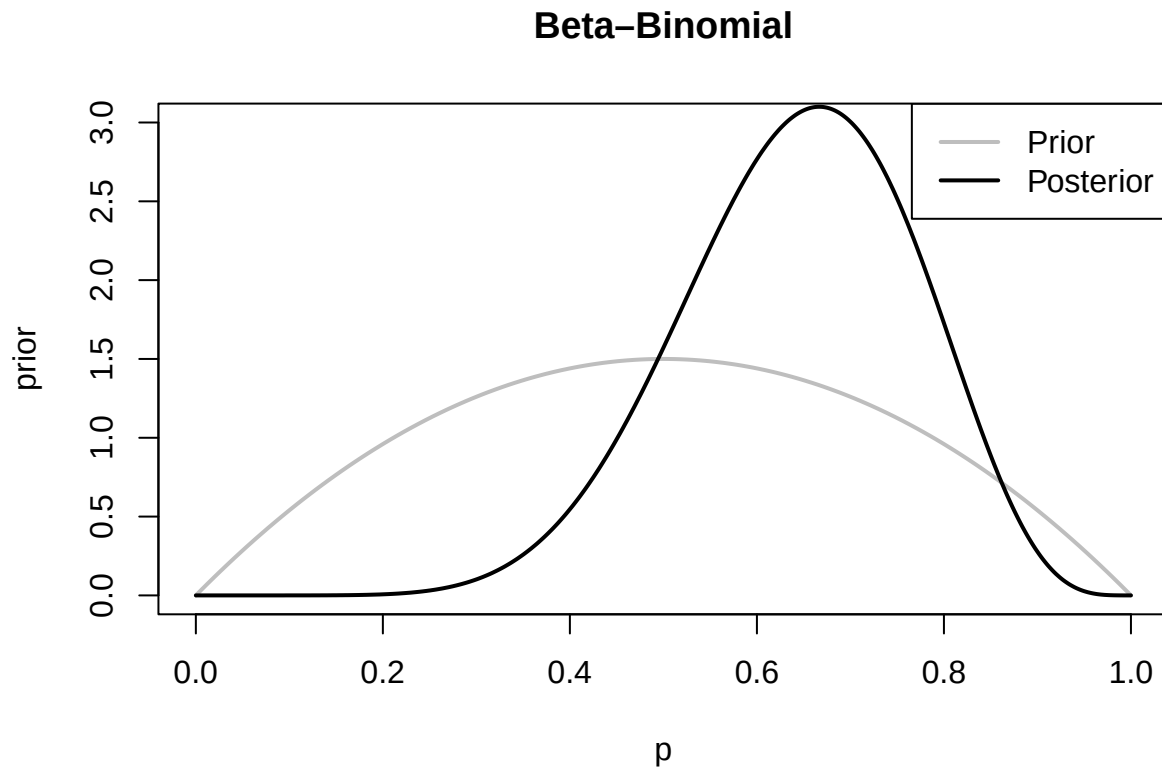
Shrinkage:

$$\text{posterior mean} = w\bar{x} + (1 - w)\text{prior mean}$$

```
x <- 7; n <- 10
alpha <- 2; beta <- 2

p <- seq(0,1,length=500)
prior <- dbeta(p, alpha, beta)
post <- dbeta(p, alpha+x, beta+n-x)

plot(p, prior, type="l", ylim=c(0,3),lwd=2, col="gray", main="Beta-Binomial")
lines(p, post, lwd=2)
legend("topright", c("Prior","Posterior"), lwd=2, col=c("gray","black"))
```



Normal-Normal model

$$X_i | \mu \sim N(\mu, \sigma^2), \text{ where } \sigma^2 \text{ is known}$$

and the prior for μ is

$$\mu \sim N(\mu_0, \tau^2)$$

What does the data tell us about μ ?

Because the X_i 's are normal,

$$\bar{X} | \mu \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

This means that the likelihood, as a function of μ , has the same shape

$$\mu | \bar{x} \propto N\left(\bar{x}, \frac{\sigma^2}{n}\right)$$

The main idea is that:

- The **prior** says that μ is near μ_0 .
- The **data** says μ is near $\bar{x}$.
- The **posterior** will be a compromise between the two.

Normal-Normal Model

$$\pi(\mu | x) \propto \underbrace{N\left(\mu | \bar{x}, \frac{\sigma^2}{n}\right)}_{\text{likelihood as a function of } \mu} \times \underbrace{N(\mu | \mu_0, \tau^2)}_{\text{prior}}.$$

Use the identity:

$$N(\mu | m_1, v_1) N(\mu | m_2, v_2) \propto N\left(\mu \left| \frac{m_1/v_1 + m_2/v_2}{1/v_1 + 1/v_2}, \frac{1}{1/v_1 + 1/v_2} \right.\right).$$

Applying this with

$$m_1 = \bar{x}, \quad v_1 = \frac{\sigma^2}{n}, \quad m_2 = \mu_0, \quad v_2 = \tau^2,$$

we obtain

$$\mu | x \sim N(\mu_n, \tau_n^2),$$

where

$$\tau_n^2 = \frac{1}{n/\sigma^2 + 1/\tau^2}$$

and

$$\mu_n = \frac{n\bar{x}/\sigma^2 + \mu_0/\tau^2}{n/\sigma^2 + 1/\tau^2}.$$

Posterior = product of Gaussians $\Rightarrow$ Gaussian with added precisions

This is why precision appears: multiplying two quadratic exponentials adds their coefficients.

Interpretation:

- Posterior mean = precision-weighted average

- Precision = inverse variance
- More data $\rightarrow$ posterior variance shrinks
- In the limit as $n \rightarrow \infty$, the influence of the prior vanishes

Precision is how certain we are about a value. Higher precision means less variability (less spread, more confidence).

$$\text{Precision} = \frac{1}{\text{Variance}}$$

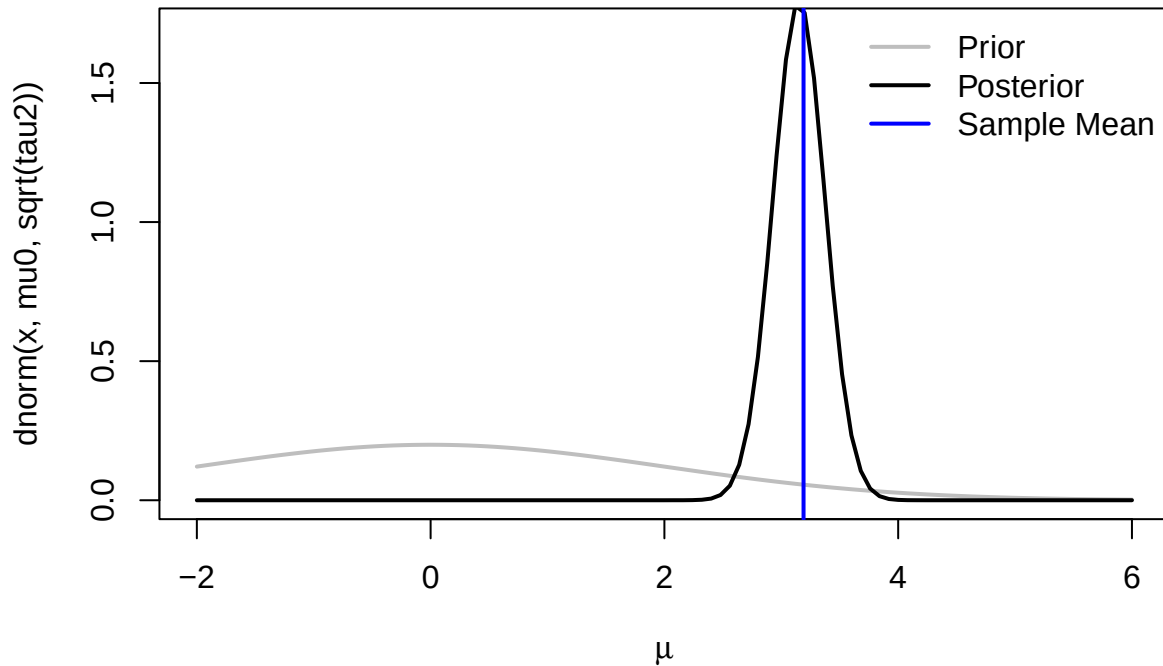
```
set.seed(1)
x <- rnorm(20, mean=3, sd=1)

mu0 <- 0; tau2 <- 4
sigma2 <- 1

tau_n2 <- 1 / (length(x)/sigma2 + 1/tau2)
mu_n <- tau_n2 * (length(x)*mean(x)/sigma2 + mu0/tau2)

curve(dnorm(x, mu0, sqrt(tau2)), -2, 6, xlab=expression(mu), ylim=c(0,1.7), lwd=2, col="gray",
      main="Normal-Normal Updating")
curve(dnorm(x, mu_n, sqrt(tau_n2)), add=TRUE, lwd=2)
abline(v = mean(x), lwd = 2, col = "blue")
legend("topright",
      c("Prior", "Posterior", "Sample Mean"),
      lwd=2,
      col=c("gray", "black", "blue"),
      bty="n")
```

Normal–Normal Updating



Conjugate priors are used because they make Bayesian updating transparent and interpretable.

Why flat priors fail

In the previous lecture, we used a constant function $\text{Uniform}(0, 1)$ as a prior. This is often *called* a non-informative prior, although this terminology can be misleading.

The idea is that we have no strong prior information favoring any particular parameter value, so we assign equal weight across the parameter space.

Suppose $X \sim N(\mu, \sigma^2)$ with σ known. If we use a flat prior $\pi(\mu) = c$, $c > 0$, then since the normal distribution has support $(-\infty, \infty)$, the prior must also have support on the entire real line. In this case, $\pi(\mu)$ cannot be normalized and therefore is not a valid probability density function.

Such a prior is called an *improper prior*. Although improper priors do not integrate to one, they can still lead to proper posterior distributions.

Another issue with flat priors is that they are not invariant under reparameterization.

Suppose $X \sim \text{Bernoulli}(p)$ and we use a flat prior $\pi(p) = 1$ on $(0, 1)$. Now define a transformed parameter

$$\psi = \log \left(\frac{p}{1-p} \right).$$

The induced density of ψ is

$$f_{\Psi}(\psi) = \frac{e^{\psi}}{(1 + e^{\psi})^2},$$

which is not flat. Thus, a flat prior on p does not correspond to a flat prior on ψ .

How did the density $f_{\Psi}(\psi)$ come about?

$$\psi = \log\left(\frac{p}{1-p}\right)$$

$$e^{\psi} = \frac{p}{1-p}$$

$$p = \frac{e^{\psi}}{1 + e^{\psi}}$$

The transformation theorem says,

$$f_{\Psi}(\psi) = f_p(p) \left| \frac{dp}{d\psi} \right|$$

Now, assume $p \sim \text{Unif}(0, 1)$, so $f_p(p) = 1$ (i.e. no prior information on p).

$$\frac{dp}{d\psi} = \frac{e^{\psi}(1 + e^{\psi}) - e^{\psi}e^{\psi}}{(1 + e^{\psi})^2} = \frac{e^{\psi}}{(1 + e^{\psi})^2}$$

$$f_{\Psi}(\psi) = (1) \frac{e^{\psi}}{(1 + e^{\psi})^2} = \frac{e^{\psi}}{(1 + e^{\psi})^2}$$

If we are truly ignorant about p , then we should also be ignorant about any one-to-one transformation of p . This shows that flat priors are not parameterization invariant.

Jeffreys Prior

Jeffreys (1891–1989) proposed a rule for constructing priors that are invariant under reparameterization. The Jeffreys prior is defined as

$$\pi(\theta) \propto \sqrt{I(\theta)},$$

where $I(\theta)$ is the Fisher information.

This choice yields a prior that is invariant under one-to-one transformations of the parameter. Thus, Jeffreys priors formalize the idea of ‘ignorance’ in a parameterization-invariant way

Example.

Consider the Bernoulli(p) model. The Fisher information is

$$I(p) = \frac{1}{p(1-p)}.$$

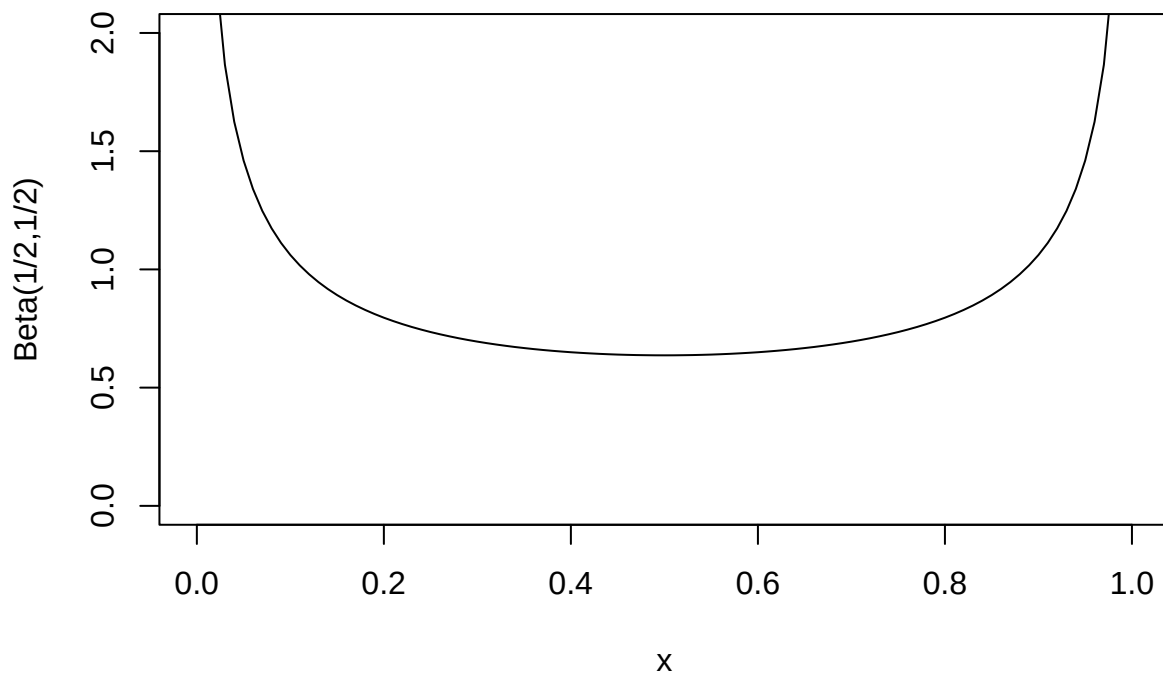
The corresponding Jeffreys prior is

$$\pi(p) \propto \sqrt{\frac{1}{p(1-p)}} = p^{-1/2}(1-p)^{-1/2},$$

which is a Beta(1/2, 1/2) distribution.

This prior places relatively little structure in the interior of (0, 1) while assigning more weight near the boundaries $p = 0$ and $p = 1$.

This is a Beta(1/2, 1/2) density, which is symmetric but places more weight near the boundaries $p = 0$ and $p = 1$



Example: Normal Mean with Known Variance

Suppose,

$$X \mid \mu \sim N(\mu, \sigma^2),$$

where σ^2 is known and $\mu \in (-\infty, \infty)$.

The log-likelihood for a single observation is

$$\ell(\mu) = -\frac{1}{2\sigma^2}(x - \mu)^2 + \text{const.}$$

Differentiating twice with respect to μ ,

$$\frac{\partial^2 \ell(\mu)}{\partial \mu^2} = -\frac{1}{\sigma^2}.$$

Thus, the Fisher information is

$$I(\mu) = -\mathbb{E}\left[\frac{\partial^2 \ell(\mu)}{\partial \mu^2}\right] = \frac{1}{\sigma^2}.$$

Since $I(\mu)$ does not depend on μ , the Jeffreys prior is

$$\pi(\mu) \propto \sqrt{I(\mu)} = \sqrt{\frac{1}{\sigma^2}} \propto 1.$$

Therefore, the Jeffreys prior for the Normal mean (with known variance) is a constant prior on $(-\infty, \infty)$.

Remark.

This prior is improper, but it is invariant under reparameterization and leads to a proper posterior distribution when combined with the Normal likelihood.

Summary

Big Takeaways

Bayesian updating = prior information + data information

Posterior parameters are obtained by adding information: + Beta–Binomial: counts add
 + Normal–Normal: precisions add
 + General case: natural parameters add

Bayesian updating = prior information + data information

- The posterior is a compromise between prior beliefs and data, weighted by information.
- As $n \rightarrow \infty$, the data dominates the prior.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

# Data
x = 7
n = 10
alpha = 2
beta_param = 2 # avoid shadowing scipy.stats.beta

# Grid
p = np.linspace(0, 1, 500)

# Prior and posterior densities
prior = beta.pdf(p, alpha, beta_param)
post = beta.pdf(p, alpha + x, beta_param + n - x)
```

```

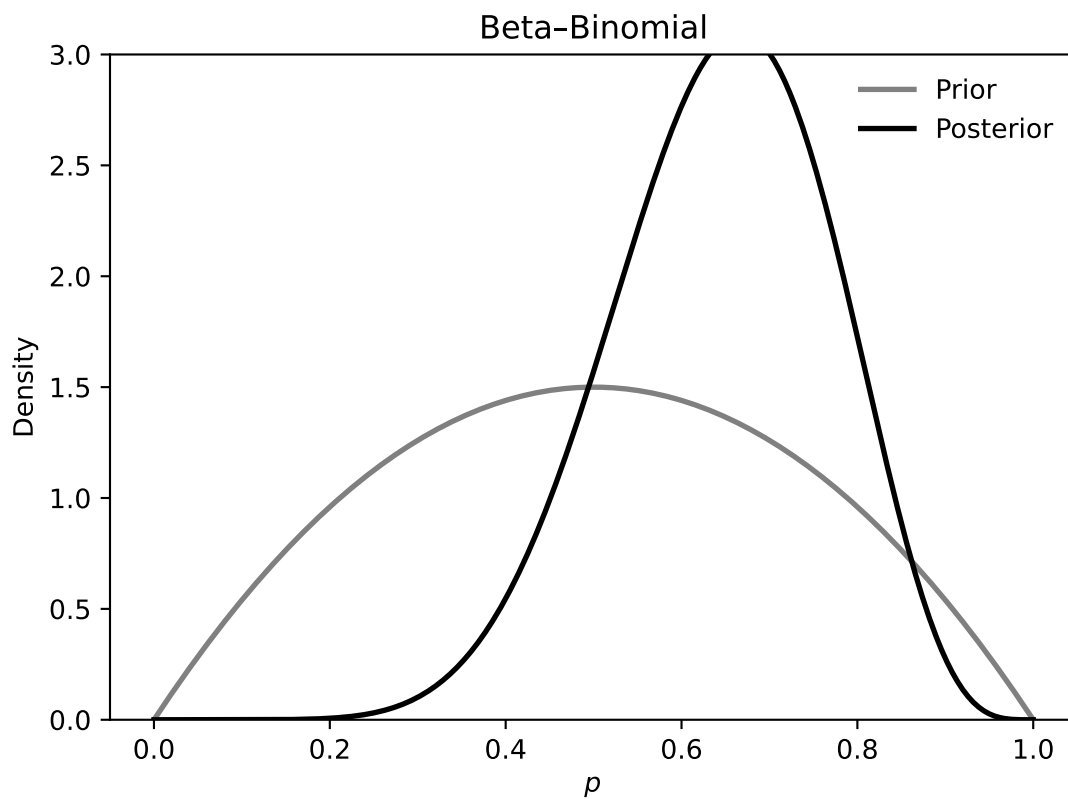
# Plot
plt.figure()
plt.plot(p, prior, linewidth=2, color="gray", label="Prior")
plt.plot(p, post, linewidth=2, color="black", label="Posterior")

plt.ylim(0, 3)

## (0.0, 3.0)
plt.title("Beta-Binomial")
plt.xlabel(r"$p$")
plt.ylabel("Density")
plt.legend(loc="upper right", frameon=False)

plt.show()

```



```

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Seed (matches set.seed(1) as closely as possible)
np.random.seed(1)

# Data
x = np.random.normal(loc=3, scale=1, size=20)

```

```

mu0 = 0
tau2 = 4
sigma2 = 1

n = len(x)

# Posterior parameters
tau_n2 = 1.0 / (n / sigma2 + 1.0 / tau2)
mu_n = tau_n2 * (n * np.mean(x) / sigma2 + mu0 / tau2)

# Grid for plotting
mu = np.linspace(-2, 6, 500)

# Prior and posterior densities
prior = norm.pdf(mu, mu0, np.sqrt(tau2))
post = norm.pdf(mu, mu_n, np.sqrt(tau_n2))

# Plot
plt.figure()
plt.plot(mu, prior, linewidth=2, color="gray", label="Prior")
plt.plot(mu, post, linewidth=2, color="black", label="Posterior")
plt.axvline(np.mean(x), linewidth=2, color="blue", label="Sample Mean")

plt.xlim(-2, 6)

## (-2.0, 6.0)

plt.ylim(0, 1.7)

## (0.0, 1.7)

plt.xlabel(r"$\mu$")
plt.ylabel("Density")
plt.title("Normal-Normal Updating")
plt.legend(loc="upper right", frameon=False)

plt.show()

```

